

Random Signal Theory–EEE 554

Lecture Notes 1

Review on Basic Probability

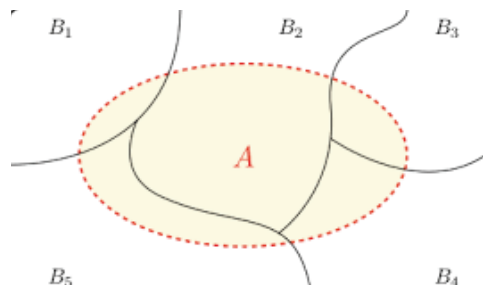
[Reading: Kay, Sections 3.3-3.9, 4.3-4.5]

Law of total probability

Consider a *partitioning* of the sample space $\mathcal{S}$ into N events $\mathcal{B}_i$, $i = 1, \dots, N$

Any two events in a *partition* do not intersect (events are mutually exclusively)
all events together form the sample space; specifically

$\mathcal{B}_i \cap \mathcal{B}_j = \emptyset$, for $i \neq j$, $i, j = 1, \dots, N$, and $\mathcal{S} = \mathcal{B}_1 \cup \mathcal{B}_2 \cup \dots \cup \mathcal{B}_N$



Given an arbitrary event $\mathcal{A}$ in the sample space

$$\Pr(\mathcal{A}) = \sum_{i=1}^N \Pr(\mathcal{A} \cap \mathcal{B}_i) = \sum_{i=1}^N \Pr(\mathcal{A} | \mathcal{B}_i) \Pr(\mathcal{B}_i)$$

► From Bayes' theorem:

$$\Pr(\mathcal{B}_i | \mathcal{A}) = \frac{\Pr(\mathcal{A} | \mathcal{B}_i) \Pr(\mathcal{B}_i)}{\Pr(\mathcal{A})} = \frac{\Pr(\mathcal{A} | \mathcal{B}_i) \Pr(\mathcal{B}_i)}{\sum_{j=1}^N \Pr(\mathcal{A} | \mathcal{B}_j) \Pr(\mathcal{B}_j)}$$

Independent Events

- ▶ Two events $\mathcal{A}$ and $\mathcal{B}$ are independent if $\Pr(\mathcal{A} \mathcal{B}) = \Pr(\mathcal{A}) \Pr(\mathcal{B})$

$\mathcal{A}$ occurs independently of whether $\mathcal{B}$ occur; given $\mathcal{B}$ does not change $\mathcal{A}$

$$\Pr(\mathcal{A} | \mathcal{B}) = \frac{\Pr(\mathcal{A} \mathcal{B})}{\Pr(\mathcal{B})} = \frac{\Pr(\mathcal{A}) \Pr(\mathcal{B})}{\Pr(\mathcal{B})} = \Pr(\mathcal{A})$$

- ▶ If $\mathcal{A}$ and $\mathcal{B}$ are independent, then $\mathcal{A}^c$ and $\mathcal{B}$ are also independent
- ▶ If $\mathcal{B}^c$ and $\mathcal{A}$ are independent, then $\mathcal{A}^c$ and $\mathcal{B}^c$ are also independent
- ▶ If $\mathcal{B}_i, i = 1, \dots, N$ are N independent events, then $\Pr(\mathcal{B}_1 \mathcal{B}_2 \dots \mathcal{B}_N) = \prod_{i=1}^N \Pr(\mathcal{B}_i)$

Also, if all N are independent, then any group of them can be shown to be independent

e.g., for $N = 6$, $\Pr(\mathcal{B}_2 \mathcal{B}_4 \mathcal{B}_5) = \Pr(\mathcal{B}_2) \Pr(\mathcal{B}_4) \Pr(\mathcal{B}_5)$

Example: Probability of error in a binary communications system

Let $X \in \{0, 1\}$ transmitted symbol and $Y \in \{0, 1\}$ received symbol

Let $\Pr(X = i \cap Y = j) = \Pr(X_i \cap Y_j) = \Pr(X_i Y_j)$, $i, j = 0, 1$

be the joint probability of $\{X = i \textbf{ and } Y = j\}$

Let $\Pr(X_0) = \Pr(X_1) = 0.5$

Let $\Pr(Y_1 | X_1) = \Pr(Y_0 | X_0) = 0.9$

and $\Pr(Y_1 | X_0) = \Pr(Y_0 | X_1) = 0.1$ (from measurements)

[Q1] To find the joint probabilities, $\Pr(X_i Y_j)$, $i, j = 0, 1$, we use Bayes' rule

$$\Pr(X_i Y_i) = \Pr(Y_i | X_i) \Pr(X_i) = 0.9(0.5) = 0.45, \quad i = 0, 1$$

$$\Pr(X_i Y_j) = \Pr(Y_j | X_i) \Pr(X_i) = 0.1(0.5) = 0.05, \quad i = 0, j = 1 \text{ and } i = 1, j = 0$$

Intersection probabilities provided in tabular form

	X_0	X_1
Y_0	0.45	0.05
Y_1	0.05	0.45

referring to the first row (first column) in the table as the zeroth row (zeroth column),
 adding the elements of the i th row yields $\Pr(Y_i)$

adding the elements of the i th column yields $\Pr(X_i)$

$$\Pr(X_1) = \Pr(Y_0 X_1) + \Pr(Y_1 X_1)$$

[Q2] To find the probabilities of the received symbols, we use total probability theorem

$$\Pr(Y_i) = \sum_{j=0}^1 \Pr(Y_i | X_j) \Pr(X_j), \quad i = 0, 1$$

So, $\Pr(Y_0) = (0.9)(0.5) + (0.1)(0.5) = 0.5$ and similarly, $\Pr(Y_1) = 0.5$

[Q3] To find the conditional probabilities $\Pr(X_i | Y_j)$, $i, j = 0, 1$, we use Bayes' rule

$$\Pr(X_i | Y_i) = \frac{\Pr(X_i Y_i)}{\Pr(Y_i)} = 0.45/0.5 = 0.9, \quad i = 0, 1$$

$$\Pr(X_j | Y_i) = \frac{\Pr(X_j Y_i)}{\Pr(Y_i)} = 0.05/0.5 = 0.1, \quad i = 0, j = 1 \text{ and } i = 1, j = 0$$

[Q4] To find the probability of error, find $\Pr(\mathcal{A})$, where event

$\mathcal{A} = \{ \text{"receive 0 and 1 is transmitted"} \text{ or } \text{"receive 1 and 0 is transmitted"} \text{ or both} \}$

$\Rightarrow \mathcal{A} = \{X_1 Y_0 + X_0 Y_1\}$ since $X_1 Y_0$ and $X_0 Y_1$ are two mutually exclusive events

Probability of error $P_e = \Pr(\mathcal{A})$ is given by

$$\begin{aligned} P_e &= \Pr(X_1 Y_0) + \Pr(X_0 Y_1) \\ &= \Pr(Y_0 | X_1) \Pr(X_1) + \Pr(Y_1 | X_0) \Pr(X_0) = 0.05 + 0.05 = 0.1 \end{aligned}$$

Example demonstrates importance of Bayes' rule, used to find information about events such as $X_1 Y_0$ or $X_1 | Y_0$ that are not observed directly